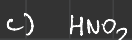
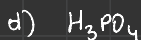


Oblicz pH roztworu kwasu :



$$K_a = 5,62 \cdot 10^{-4}$$



$$K_{a1} = 6,92 \cdot 10^{-3}$$

o stężeniu $0,1 \text{ mol/dm}^3$



$$[\text{H}^+] = 0,1 \text{ mol/dm}^3$$

$$\text{pH} = -\log [\text{H}^+] = -\log 0,1 = 1$$



$$[\text{H}^+] = 0,2 \text{ mol/dm}^3$$

$$\text{pH} = -\log 0,2 = 0,7$$



$$K_a = 5,62 \cdot 10^{-4}$$

c_0	0,1	0	0
Δc	-x	+x	+x
c_k	0,1-x	x	x

$$K_a = \frac{[\text{H}^+] \cdot [\text{NO}_2^-]}{[\text{HNO}_2]}$$

$$5,62 \cdot 10^{-4} = \frac{x \cdot x}{0,1-x}$$

$$\text{pH} = -\log (7,22 \cdot 10^{-3})$$

$$5,62 \cdot 10^{-4} (0,1-x) = x^2 \quad = 2,14$$

$$x_1 = \frac{-5,62 \cdot 10^{-4} - 0,015}{2} < 0$$

$$x^2 + 5,62 \cdot 10^{-4} x - 5,62 \cdot 10^{-5} = 0$$

$$x_2 = \frac{-5,62 \cdot 10^{-4} + 0,015}{2}$$

$$\Delta = (5,62 \cdot 10^{-4})^2 - 4 \cdot 1 \cdot (-5,62 \cdot 10^{-5})$$
$$= 2,25 \cdot 10^{-4}$$

$$= 7,22 \cdot 10^{-3}$$

$$\sqrt{\Delta} = 0,015$$

$$K = \frac{C_0 \cdot \alpha^2}{1 - \alpha}$$

C_0 - stężenie

α - stopień dysocjacji
(wydajność reakcji dysocjacji)

$$\alpha < 0,05 \quad \frac{C_0}{K} > 400$$

$$K = C_0 \cdot \alpha^2$$

sprawdzamy, czy można zastosować upr.

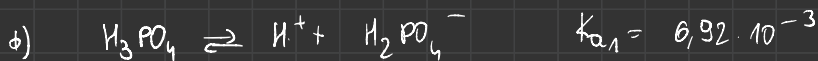
$$\frac{C_0}{K} = \frac{0,1}{5,62 \cdot 10^{-4}} = 178 < 400$$

$$5,62 \cdot 10^{-4} = \frac{0,1 \cdot \alpha^2}{1 - \alpha} \quad | \cdot (1 - \alpha)$$

$$5,62 \cdot 10^{-4} - 5,62 \cdot 10^{-4} \alpha = 0,1 \alpha^2$$

$$0,1 \alpha^2 + 5,62 \cdot 10^{-4} \alpha - 5,62 \cdot 10^{-4} = 0$$

$$\alpha = \frac{C_{\text{zd}}}{C_0} = \frac{[H^+]}{C_0}$$



C_0	0,1	0	0
ΔC	-x	+x	+x
C_K	0,1-x	x	x

$$6,92 \cdot 10^{-3} = \frac{x^2}{0,1-x}$$

$$6,92 \cdot 10^{-3} (0,1-x) = x^2$$

$$-x^2 - 6,92 \cdot 10^{-3} x + 6,92 \cdot 10^{-4} = 0$$

$$x_1 = \frac{6,92 \cdot 10^{-3} - 0,053}{-2}$$

$$\Delta = (-6,92 \cdot 10^{-3})^2 - 4 \cdot (-1) \cdot 6,92 \cdot 10^{-4} = 2,82 \cdot 10^{-3}$$

$$= 0,023$$

$$\sqrt{\Delta} = 0,053$$

$$\text{pH} = -\log 0,023 = 1,64$$

Jaki będzie odczyn roztworu:

(roztwór o stężeniu $0,1 \text{ mol/dm}^3$)

a) chlorku amonu $\begin{matrix} \text{sz} & \text{mk} \\ \text{NH}_4\text{Cl} \end{matrix}$

kwasowy

b) chlorku sodu $\begin{matrix} \text{mz} & \text{mk} \\ \text{NaCl} \end{matrix}$

obojętny

c) fosforanu(V) sodu $\begin{matrix} \text{mz} & \text{sz} \\ \text{Na}_3\text{PO}_4 \end{matrix}$

zasadowy

d) azotanu(III) amonu NH_4NO_2

(lekko) kwasowy

$$\text{NH}_3: K_b = 1,78 \cdot 10^{-5}$$

$$\text{HNO}_2: K_a = 5,62 \cdot 10^{-4}$$

Odczyn jest skutkiem hydrolyzy jonu słabej zasady / kwasu



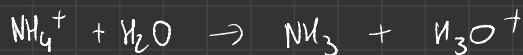
$$\text{NH}_3: K_b = 1,78 \cdot 10^{-5}$$

$$\text{NH}_4^+: * K_a = 5,62 \cdot 10^{-10}$$

$$K_a \cdot K_b = 10^{-14}$$

$$K_a = \frac{10^{-14}}{K_b} = \frac{10^{-14}}{1,78 \cdot 10^{-5}}$$

$$= 5,62 \cdot 10^{-10} *$$



c_0	0,1	\	0	0
Δc	-x	\	+x	+x
c_K	0,1-x	\	x	x

$$\frac{0,1}{5,62 \cdot 10^{-10}} = 178\,000\,000 > 400$$

$$\frac{c}{K} > 400$$

$$5,62 \cdot 10^{-10} = \frac{x^2}{0,1-x}$$

Wb

$$K = c \cdot \alpha^2$$

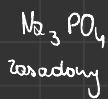
$$\alpha = \sqrt{\frac{K}{c}}$$

$$\alpha = \sqrt{\frac{5,62 \cdot 10^{-10}}{0,1}} = 7,5 \cdot 10^{-5}$$

$$\alpha = \frac{c_{\text{ad.}}}{c_0} = \frac{[\text{H}_3\text{O}^+]}{c_0}$$

$$\text{pH} = -\log(7,5 \cdot 10^{-6}) = 5,12$$

$$[\text{H}_3\text{O}^+] = \alpha \cdot c_0 = 7,5 \cdot 10^{-5} \cdot 0,1 = 7,5 \cdot 10^{-6} \text{ mol/dm}^3$$



$$K_{a3} = 4,73 \cdot 10^{-13} \rightarrow$$



$$K_{b1} =$$

$$K_a = \frac{[\text{PO}_4^{3-}] \cdot [\text{H}_3\text{O}^+]}{[\text{HPO}_4^{2-}]}$$

$$K_b = \frac{[\text{HPO}_4^{2-}] \cdot [\text{OH}^-]}{[\text{PO}_4^{3-}]}$$

$$K_a \cdot K_b = \frac{[\text{PO}_4^{3-}] \cdot [\text{H}_3\text{O}^+]}{[\text{HPO}_4^{2-}]} \cdot \frac{[\text{HPO}_4^{2-}] \cdot [\text{OH}^-]}{[\text{PO}_4^{3-}]} = [\text{H}_3\text{O}^+] \cdot [\text{OH}^-] = 10^{-14} \quad T=25^\circ\text{C}$$

$$K_a \cdot K_b = 10^{-14}$$

$$K_b = \frac{10^{-14}}{4,73 \cdot 10^{-13}} = 0,0209$$



c_0	0,1	\	0	0
Δc	-x	\	+x	+x
c_k	0,1-x	\	x	x

$$\frac{x^2}{0,1-x} = 0,0209$$

$$[\text{OH}^-] = 0,036 \frac{\text{mol}}{\text{dm}^3}$$

$$-x^2 - 0,0209x + 0,00209 = 0$$

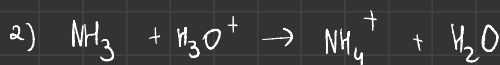
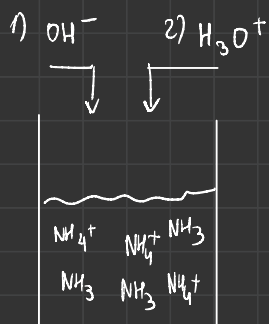
$$\text{pOH} = -\log 0,036 = 1,44$$

$$\sqrt{\Delta} = 0,0338$$

$$\text{pH} = 14 - 1,44 = \underline{\underline{12,56}}$$

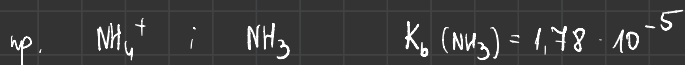
$$x_1 = \frac{0,0209 - 0,0338}{-2} = 0,036$$

Bufor pH zmienia ułamek ilości substancji kwasowej i sprzężonej zasady kwasu



Bufor nie zmienia znacząco pH po dodaniu kwasu (H_3O^+) lub zasady (OH^-).

pH buforu



$$K_b = \frac{[\text{NH}_4^+] \cdot [\text{OH}^-]}{[\text{NH}_3]}$$

$K_b \cdot \frac{[\text{NH}_3]}{[\text{NH}_4^+]} = [\text{OH}^-] \quad | \quad \log$

$\log \left(K_b \cdot \frac{[\text{NH}_3]}{[\text{NH}_4^+]} \right) = \log [\text{OH}^-]$

Jak uzyskać pH?

$\text{pH} = -\log [\text{H}^+]$

$\text{pH} = 14 - \text{pOH}$

$\text{pH} = 14 - \log [\text{OH}^-]$

$\text{pH} = 14 - \log \left(K_b \cdot \frac{[\text{NH}_3]}{[\text{NH}_4^+]} \right)$

$\text{pH} = 14 - \left(\log K_b + \log \frac{[\text{NH}_3]}{[\text{NH}_4^+]} \right)$

$\text{pH} = 14 + \text{p}K_b - \log \frac{[\text{NH}_3]}{[\text{NH}_4^+]}$

$\text{pH} = 14 - \text{p}K_b - \log \frac{[\text{NH}_3]}{[\text{NH}_4^+]}$

$$pH = pK_a + \log \frac{[A^-]}{[HA]}$$

W roztworze znajduje się 0,4 mol usteceku CH_3COOH i 0,4 mol anionów CH_3COO^- . Oblicz pH takiego roztworu buforowego.

$$pK_a = 4,756$$

$$pH = 4,756 + \log \left(\frac{\frac{0,4}{V}}{\frac{0,4}{V}} \right)$$

$$pH = 4,756 + \log_{10} 1 = 4,756 + 0 = 4,756$$

$$10^x = 1$$